VoiceOps 2026 Capstone Report

ZentroMart Customer Support Voice Agent

Project overview

This project implements a customer-support agent for the fictional retailer ZentroMart. It combines a Gemini-powered conversational agent, retrieval-augmented generation (RAG) over the supplied company knowledge base, synthetic order lookup and ticket-creation tools, and Gnani speech services. The implementation uses the supplied capstone dataset and company documents.

Technology

Python; Google Gemini (gemini-3.5-flash-lite); Sentence Transformers (all-MiniLM-L6-v2); FAISS; pandas; python-dotenv; Gnani Vachana SDK. Voice settings: en-IN, Kaveri, timbre-v2.5.

1. What was built in each step

Step 1 -Basic chatbot: A Gemini chatbot was created with a customer-support system prompt, providing the initial text-based baseline.

Step 2-RAG: Supplied Markdown knowledge-base documents were divided into sections, embedded with all-MiniLM-L6-v2, and indexed in FAISS. The index contains 84 sections. At runtime, relevant sections are retrieved and passed to Gemini as context. Return-policy tests retrieved current policy sections, including pol_returns_v4.md; an electronics return-window test returned seven calendar days from the RET-3 section.

Step 3-Multi-turn agent and tools: The agent maintains conversation state, asks for an order ID, verifies the caller using the registered phone suffix or delivery pincode, and uses the supplied lookup_order and create_ticket tools. Escalation rules cover human-agent requests, repeat contacts, unauthorised activity, legal threats, product-safety risks, high-value damaged items, and overdue refunds. Ticket creation is confirmed only when the tool reports success.

Step 4-Voice: Gnani STT transcribes customer audio and Gnani TTS speaks the reply. The voice runner accepts one audio file or a folder and saves reply text, reply audio, and retrieved-chunk JSON for each input.

2. Testing and observed results

Functional checks observed during development:

1)RAG returned relevant current-policy sections for return-policy questions.

2)Multi-turn order flow requested an eight-digit order ID, converted spoken digit words from an ASR transcription into digits, and then requested identity verification before disclosing order details.

3)Tool and escalation tests covered a human request, repeat contact, unauthorised payment activity, legal action, product safety, a high-value damaged item, and an overdue cancellation refund. Successful tool calls returned ticket IDs.

4)Gnani STT and TTS were exercised on recorded audio. Batch processing handled two audio files and saved reply.wav, reply.txt, and retrieved_chunks.json for each.

Step 1 vs Step 2 scores

A formal numeric comparison across all 52 practice questions was not recorded in the evidence available for this report. No

scores are invented. Qualitatively, Step 1 relied on the model prompt, while Step 2 supplied retrieved policy sections to ground answers. Insert measured baseline and RAG scores if the practice evaluation is completed before submission.

3. What went wrong and what was improved

- Spoken order numbers were initially transcribed as English number words, but the agent accepted only numeric characters. Order-ID handling was updated to recognise sequences such as "three nine one ... " and convert them to digits; the multi-turn test then advanced to identity verification.
- An early voice batch run used an outdated single-file version. The batch-capable script was installed and verified to accept a folder.
- Separate output folders are used per audio input so text, speech, and retrieved evidence remain together.

4. Limitations and future improvements

The formal 52-question Step 1 versus Step 2 score sheet has not yet been produced, so quantitative gains are not claimed. The development batch test covered two audio files; a full evaluation should process the complete evaluation audio set supplied by the organisers. Further improvements include a repeatable evaluation script, broader language and noisy-audio testing, and a clean-environment installation test.

5. Reproducibility

The project includes requirements.txt, a key-free .env.example, and run.py. Intended batch command: python run.py customer_audio. Set GEMINI_API_KEY and GNANI_API_KEY in a private .env file. Do not include .env, API keys, the virtual environment, or a copy of the supplied dataset in the submission ZIP